Mrugank Dake

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Hanover, NH

ACADEMIC HISTORY

Dartmouth College Hanover, NH
Postdoctoral Researcher in Psychological and Brain Sciences **June 2026 – Present**
Advisor: Dr. John Murray

New York University New York, NY
PhD in Computational Cognitive Neuroscience **May 2026**
Advisor: Dr. Clayton Curtis
Dissertation: The nature and role of distributed representation in Working Memory
MPhil in Experimental Psychology **May 2025**
Coursework: Linear Algebra for Neural & Cognitive Science, fMRI Analysis Methods, Computational Neuroscience

Indian Institute of Science Education & Research (IISER) Tirupati, India
BS-MS Dual Degree (Biology & Physics) **August 2021**
Advisor: Dr. Nandini Rajamani
Master's Thesis: Statistical and Machine learning approaches to Squirrel Acoustics
Coursework: Nonlinear Dynamics & Chaos, Statistical Thermodynamics, Linear Algebra, Multivariable Calculus

SPECIALIZED TRAINING

- **Marine Biological Laboratory (MBL)** Woods Hole, MA
  Methods in Computational Neuroscience (MCN) Course **2026**
- **Neuromatch Academy** Virtual
  Deep Learning Course **2022**

PUBLICATIONS

- **Dake M.**, Dandekar S., Curtis C. E. (2026). Neural synchrony between prefrontal and visual cortex supports visual working memory. *bioRxiv*
- **Dake M.**, Curtis C. E. (2025). Perturbing human V1 degrades the fidelity of visual working memory. *Nature Communications*.

SOFTWARE & TOOLS

- **pRF fitting toolbox** — JAX-based toolbox for population receptive field (pRF) modeling of fMRI data using the compressive spatial summation (CSS) model; fits a full volumetric dataset in ∼5 min on a single GPU. *In development.*
- **Alexαndria** — Platform integrating shared bibliographies with a collaborative activity space for research teams. *In development.* alexandriasocial.net
- **MCN Network Graph** — Interactive visualization of the research network of the MBL Methods in Computational Neuroscience cohorts. mindemory.io/projects/mcn-network
- **EEG/MEG Methods course materials** — Open tutorials and analysis notebooks for the NYU iEEG/EEG/MEG Methods PhD course (2024). github.com/mindemory/EEG_MEG_Methods

TALKS

- **Dake M.**, Dandekar S., Curtis C. E. Neural synchrony between prefrontal and visual cortices supports visual working memory; *Working Memory Symposium (Virtual)* **July 2026**
- **Dake M.**, Curtis C. E. A tale of two cortices; *NYU TELL GSAS competition* **April 2025**

- **Dake M.**, Curtis C. E. Perturbing human V1 degrades the fidelity of visual working memory; *Working Memory Symposium (Virtual)* **July 2024**

POSTERS

- **Dake M.**, Dandekar S., Curtis C. E. Working memory synchronizes oscillations within and across cortical areas; *Society for Neuroscience (SfN), San Diego CA, USA* **Nov 2025**
- **Dake M.**, Dandekar S., Curtis C. E. Working memory synchronizes oscillations in visual cortex; *Computational Cognitive Neuroscience (CCN), Amsterdam, Netherlands* **May 2025**
- **Dake M.**, Curtis C. E. Do perturbations to visual cortex impact working memory?; *In Society for Neuroscience (SfN), Washington D.C., USA* **Nov 2023**
- Khare S., Chopra S., Grover H., Vohra T., Mahmood U., Jatwani K., Kaushik D., Pandey A., Bhardwaj S., Tanwar A., Singh J., Shekhar C., **Dake M.**, Krishnamoorthy K., Kumar K. Li-Koff: To Detect and Degrade N-Nitrosamines; *Giant Jamboree (iGEM), Boston, USA* **Nov 2020**
- Amar I., Avadhani K., Bajaj M., Balasubramanian D., Bhagat S., Chutani N., **Dake M.**, Khatri U., Krishna N., Jacob M., Mal S., Mohapatra O., Pal A., Saha D., Tripathy B., Mukherjee R., Rao B.J. CoCa coli: probiotic immunotherapy against colon cancer; *Giant Jamboree (iGEM), Boston, USA* **Nov 2019**

PRE-DOCTORAL RESEARCH EXPERIENCE

- Statistical and Machine learning approaches to Squirrel Acoustics (Masters' Thesis) **Aug 2020 - Aug 2021**
  Advisor: Dr. Nandani Rajamani, IISER Tirupati, India
- EEG and MRS Analysis for underpinning neural correlates of attentional blink **June - Aug 2019**
  Advisor: Dr. Sridharan Devarajan, IISc Bangalore, India
- CoCa coli: probiotic immunotherapy against colon cancer **Jan 2018 - Nov 2019**
  Advisors: Dr. Raju Mukherjee; Dr B J Rao, IISER Tirupati, India
- Isolation and characterization of neurons and glia from goat hippocampal culture **Sep 2018 - May 2019**
  Advisor: Dr. Vasudharani Devanathan, IISER Tirupati, India
- Identifying differentially expressing tumor suppressor genes in *Drosophila melanogaster* wing and haltere imaginal discs **May - July 2018**
  Advisor: Prof. L S Shashidhara, IISER Pune, India
- RNA Origami (Square) using in-vitro replication and transcription **Jan - March 2018**
  Advisor: Dr. Ashwani Sharma, IISER Tirupati, India
- Head direction cell model for rodents with internal updating mechanism **May - July 2017**
  Advisor: Dr. Collins Assisi, IISER Pune, India

PROFESSIONAL & TEACHING EXPERIENCE

NYU Leslie Entrepreneurial Institute New York, NY
Leslie Founder **Feb 2026 – Present**

Meta Reality Labs Burlingame, CA
Product Data Science (PhD) Intern **May – Aug 2025**

New York University New York, NY
Teaching Assistant: Scientific Programming (MA Course) **Fall 2025**
Teaching Assistant: iEEG/EEG/MEG Methods (PhD Course) **Fall 2024**
Teaching Assistant: Introduction to Cognitive Neuroscience (UG) **Fall 2023**
Teaching Assistant: Math Tools III: Linear Systems (PhD Course) **Spring 2023**

PROFESSIONAL SERVICE

Ambassador: Thinking About Thinking **Oct 2026 - Present**
Ad-hoc Reviewer: Brain Structure & Function (2026), eLife, ICLR (2026), Cognitive Science Society (2025), Computational Cognitive Neuroscience (CCN 2025, 2026)
Website Design Coordinator: Climatematch Academy **Aug – Dec 2023**
Project Mentor: iGEM, Manav Rachna International Institue of Research **2020**

Student Team Leader: iGEM, IISER Tirupati **2019**

AWARDS

Society for Neuroscience Trainee Professional Development Award ($1,000)	**2025**
Dean's Award for Research Communication Excellence, NYU ($2,000)	**2025**
GSAS Conference Travel Grant, NYU	**2025**
MacCracken Doctoral Fellowship, NYU	**2021 – 2026**
Indian Biological Engineering Competition (iBEC) Grant (~$12,000)	**2020**
Indian Biological Engineering Competition (iBEC) Grant (~$12,000)	**2019**
INSPIRE Fellowship, Department of Science and Technology India	**2016 – 2021**
Regional Winner, 5th Place, Indian National Brain Bee (INBB)	**2015**

SKILLS

Programming
Python, MATLAB, R, SQL, LaTeX, CSS, JavaScript, HTML, React, NodeJS, Supabase, Vercel, Antigravity, VSCode

Data Science & ML
JAX, Jit, PyTorch, TensorFlow, Keras, Scikit-learn, Neural Networks, Time-series Analysis

Neuroscience Tools
EEG, MEG, fMRI, TMS, ECoG, Fieldtrip, MNE-Python, SimNIBS, Psychtoolbox

Other Tools & Languages
Git/GitHub, Adobe Illustrator, Adobe Photoshop
English (Fluent), Hindi, Marathi